

LOTUS ZHANG

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Human Centered Design & Engineering
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RESEARCH INTERESTS

Human-Computer Interaction; Accessibility; Human-Centered AI; Creativity Support

My research takes an interdisciplinary approach to advance AI-driven innovations that empower accessible creativity. I combine quantitative rigor, qualitative depth, participatory approaches, and system prototyping to inform technology design that truly supports diverse end users.

EDUCATION

- 09/2019 - 06/2026 **University of Washington, Seattle**
Ph.D. in Human Centered Design & Engineering
Advisor: Leah Findlater
Committee: Jacob O. Wobbrock, Julie Kientz, and Maya Cakmak
- 09/2019 - 05/2025 **University of Washington, Seattle**
M.S. in Human Centered Design & Engineering
- 01/2014 - 05/2019 **University of British Columbia, Vancouver**
B.A. in Computer Science and Psychology

PROFESSIONAL EXPERIENCE

- 07/2026 - 07/2027 **University of Michigan, Ann Arbor**
Postdoctoral Fellow *AHA Research Group*
- 09/2025 - 10/2025 **Microsoft Research, Redmond**
Usability Researcher *EPIC Lab*
- 06/2024 - 09/2024 **Google, San Francisco**
UX Researcher Internship, *Products for All*
- 06/2020 - 09/2020 **Meta, Seattle**
Quantitative UX Researcher Internship, *Community Integrity*
- 06/2018 - 09/2018 **University of Michigan, Ann Arbor**
Research Assistant *Information Interaction Lab*
- 09/2016 - 05/2018 **University of British Columbia, Vancouver**
Research Assistant *SPIN Lab*
- 01/2016 - 05/2016 **Axka Group Inc., Richmond**
UX Developer Internship

PUBLICATIONS

Peer-reviewed Full-Paper Conference Publications

- C.14 **Interface Support for Evaluating Disability Bias in AI Generated Images**
Kelly Avery Mack, Lucy Jiang, Lotus Zhang, Leah Findlater
ACM Conference on Human Factors in Computing Systems (CHI), 35 pages, April 2026.

- C.13 **VizXpress: Towards Expressive Visual Content by Blind Creators Through AI Support**
Lotus Zhang, Zhuohao (Jerry) Zhang, Gina Clepper, Franklin Mingzhe Li, Patrick Carrington, Jacob O. Wobbrock, Leah Findlater
 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 18 pages, October 2025.
- C.12 **Visual Privacy Management with Generative AI for Blind and Low-Vision People**
 Tanusree Sharma, Yu-Yun Tseng, Lotus Zhang, Ayae Ide, Kelly Avery Mack, Leah Findlater, Danna Gurari, Yang Wang
 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 21 pages, October 2025.
- C.11 **BIV-Priv-Seg: Locating Private Content in Images Taken by People With Visual Impairments**
 Yu-Yun Tseng, Tanusree Sharma, Lotus Zhang, Abigale Stangl, Leah Findlater, Yang Wang
 IEEE/CVF Winter Conference on Applications in Computer Vision (WACV), 15 pages, March 2025.
- C.10 **EditScribe: Non-Visual Image Editing with Natural Language Verification Loops**
 Ruei-Che Chang, Yuxuan Liu, Lotus Zhang, Anhong Guo
 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 19 pages, October 2024.
- C.09 **Designing Accessible Obfuscation Support for Blind Individuals' Visual Privacy Management**
Lotus Zhang, Abigale Stangl, Tanusree Sharma, Yu-Yun Tseng, Inan Xu, Danna Gurari, Yang Wang, Leah Findlater
 ACM Conference on Human Factors in Computing Systems (CHI), 27 pages, April 2024.
- C.08 **Understanding digital content creation needs of blind and low vision people**
Lotus Zhang, Simon Sun, Leah Findlater
 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 15 pages, October 2023.
Best Paper Honorable Mention
- C.07 **Disability-first design and creation of a dataset showing private visual information collected with people who are blind**
 Tanusree Sharma, Abigale Stangl, Lotus Zhang, Yu-Yun Tseng, Inan Xu, Leah Findlater, Danna Gurari, Yang Wang
 ACM Conference on Human Factors in Computing Systems (CHI), 15 pages, April 2023.
- C.06 **Understanding visual arts experiences of blind people**
 Franklin Mingzhe Li*, Lotus Zhang*, Maryam Bandukda, Abigale Stangl, Kristen Shinohara, Leah Findlater, Patrick Carrington
 ACM Conference on Human Factors in Computing Systems (CHI), 21 pages, April 2023.
 * Equal Contribution
- C.05 **Exploring interactive sound design for auditory websites**
Lotus Zhang, Jingyao Shao, Augustina Ao Liu, Lucy Jiang, Abigale Stangl, Adam Fourney, Meredith Ringel Morris, Leah Findlater
 ACM Conference on Human Factors in Computing Systems (CHI), 16 pages, April 2022.
- C.04 **Public versus private: How teens perceived teen-robot interactions in a school setting**
 Katelynn Oleson, Elin A Björling, Lotus Zhang, Heba Dwikat
 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 4 pages, March 2022.
- C.03 **MRAT: The mixed reality analytics toolkit**
 Michael Nebeling, Maximilian Speicher, Xizi Wang, Shwetha Rajaram, Brian D Hall, Zijian Xie, Alexander RE Raistrick, Michelle Aebersold, Edward G Happ, Jiayin Wang, Yanan Sun, Lotus Zhang, Leah E Ramsier, Rhea Kulkarni
 ACM Conference on Human Factors in Computing Systems (CHI), 16 pages, April 2020.
Best Paper Award
- C.02 **Input Accessibility: A Large Dataset and Summary Analysis of Age, Motor Ability and Input Performance.**
 Leah Findlater, Lotus Zhang
 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 6 pages, October 2020
Best Paper Award

- C.01 **Is it happy? Behavioural and narrative frame complexity impact perceptions of a simple furry robot's emotions**
Paul Bucci, [Lotus Zhang](#), Xi Laura Cang, Karon E MacLean
ACM Conference on Human Factors in Computing Systems (CHI), 11 pages, April 2018.

Peer-reviewed Journal Publications

- J.01 **Social Media through Voice: Synthesized Voice Qualities and Self-presentation**
[Lotus Zhang](#), Lucy Jiang, Nicole Washington, Augustina Ao Liu, Jingyao Shao, Adam Fourney, Meredith Ringel Morris, Leah Findlater
ACM : Human-Computer Interaction, Computer Supported Cooperative Work and Social Computing (CSCW 2021), 21 pages, April 2021.

Posters, Workshops, and Other Reviewed Publications

- P.05 **The Accessibility, Security, and Privacy Nexus: Trends and Opportunities**
Kelly Avery Mack, Yu-Jie Chen, [Lotus Zhang](#), Danna Gurari, Tanusree Sharma, Yang Wang, Leah Findlater
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), October 2025.
- P.04 **Creation, Critique, and Consumption: Exploring Generative AI Descriptions for Supporting Blind and Low Vision Professionals with Visual Tasks**
Lucy Jiang, [Lotus Zhang](#), Leah Findlater
Workshop Paper. ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), October 2025.
- P.03 **Designing accessible content creation support with blind and low vision creators**
[Lotus Zhang](#)
Doctoral Consortium. ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), October 2023.
- P.02 **Bridging the gap: Towards advancing privacy and accessibility**
Rahaf Alharbi, Robin N Brewer, Gesu India, [Lotus Zhang](#), Leah Findlater, Yixin Zou, Abigale Stangl
Accepted Workshop Proposal. ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), October 2023.
- P.01 **Infusing cuddlebits with emotion: Build your own and tell us about it**
[Lotus Zhang](#), Paul Bucci, Xi Laura Cang, Karon MacLean
Extended Abstracts. ACM Conference on Human Factors in Computing Systems (CHI), April 2018.

AWARDS & HONORS

- 2023 **Best Paper Honorable Mention**
Publication [C.08]: ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)
- 2020 **Best Paper Award**
Publication [C.02]: ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)
- 2020 **Best Paper Award**
Publication [C.03]: ACM Conference on Human Factors in Computing Systems (CHI)
- 2017 **John Alexander McDonald Scholarship in Humanities**
- 2017 **UBC Computer Science Student Service Award**
- 2014 **UBC Outstanding International Student Award**

TEACHING

University of Washington

- 2021 **Teaching Assistant** - Experimental Research Methods
- 2020 **Co-Instructor** - Directed Research Group: Current Trend in Accessibility Research

University of British Columbia

- 2018 **Teaching Assistant** - Computer Graphics
- 2017 **Teaching Assistant** - Introduction to HCI Methods

Mentoring

- 2019 **Augustina Liu** - Now senior product designer at Toast
- 2019 **Jingyao Shao** - Now MS student at New York University
- 2019 **Lucy Jiang** - Now PhD student at UW
- 2022 **Simon Sun** - Now product designer at Jama Software
- 2026 **Anh Hua** - HCDE MS student

COMMUNITY SERVICE

Reviewer

ACM CSCW: 2021
ACM CHI: 2023, 2024*, 2025, 2026*
ACM IDC: 2023
ACM HRI: 2022
ACM SIGACCESS: 2023, 2024

*Special Recognition for Outstanding Review

- 2021 **DUB Student Coordinator**
- 2018 **Student Volunteer: ACM CHI**

INVITED TALKS

- 2025 **Best-Only Scaling: An Accessible Alternative to MaxDiff Surveys**
Quant UX Conference 2025 Worldwide
- 2025 **VizXpress: Towards Expressive Visual Content by Blind Creators Through AI Support**
Paper Presentation for ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)
- 2025 **Expressive Visual Content by Blind Creators**
UW CREATE Project Showcase
- 2024 **Designing Accessible Obfuscation Support for Blind Individuals' Visual Privacy Management**
Paper Presentation for ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)
- 2023 **Designing Accessible Co-creative Tools with Blind and Low Vision Creators**
Doctoral Consortium Talk at ASSETS 2023
- 2023 **Accessibility Research & Blind Creativity**
Introduction to HCI guest lecture at University of Toronto
- 2022 **Exploring Interactive Sound Design for Auditory Websites**
Paper Presentation for ACM CHI

